

1. About Initializing Weights:

A. the reasons to randomly initialize **input weights** is to break the symmetry.

if all perceptrons start with the same initial **input weights**,

the outputs of all perceptrons in the same layer will be identical,

and this state will lead to the **input weights** adjustment of all perceptrons in the same layer will be identical during backpropagation, too.

finally, we don't get benefit from setting multiple perceptron in the same layer

B. **bias weight** doesn't need to be randomly initialized because it is sufficient to randomly initialize **input weights** to break the symmetry.

initializing **bias weight** to 0 is a common strategy.